

Name TYear 8
Investigation 2Baldivis
Secondary College

Linear Models

13840

The school is planning a disco for all the year 9 students. It is your job to organize the music for the night. You have found the following 3 alternatives for the music.

Choice 1 – Rick's Rock Group

Rick charges \$100 per hour and but charges a flat fee of \$20 for coming out

Choice 2 – Magic Mark DJ service

Mark charges \$40 per hour plus a \$200 flat fee.

Choice 3 – Steve's Death Metal Band.

Steve charges \$60 per hour plus a \$100 flat fee.

1. Complete the table of values for each option and write a rule using "C" for cost and "N" for Number of hours that could be used for each option. (6 marks)

8 marks

Rick's Rock Group

N (number of hours)	0	1	2	3	4	5	6
C (cost in \$)	20	120	220	320	420	520	620

Now write a rule you can use to calculate the cost of Rick's Rock Group.

$$C = 100n + 20$$

Magic Mark's DJ Service.

N (number of hours)	0	1	2	3	4	5	6
C (cost in \$)	200	240	280	320	360	400	440

Now write a rule you can use to calculate the cost Magic Mark's DJ Service.

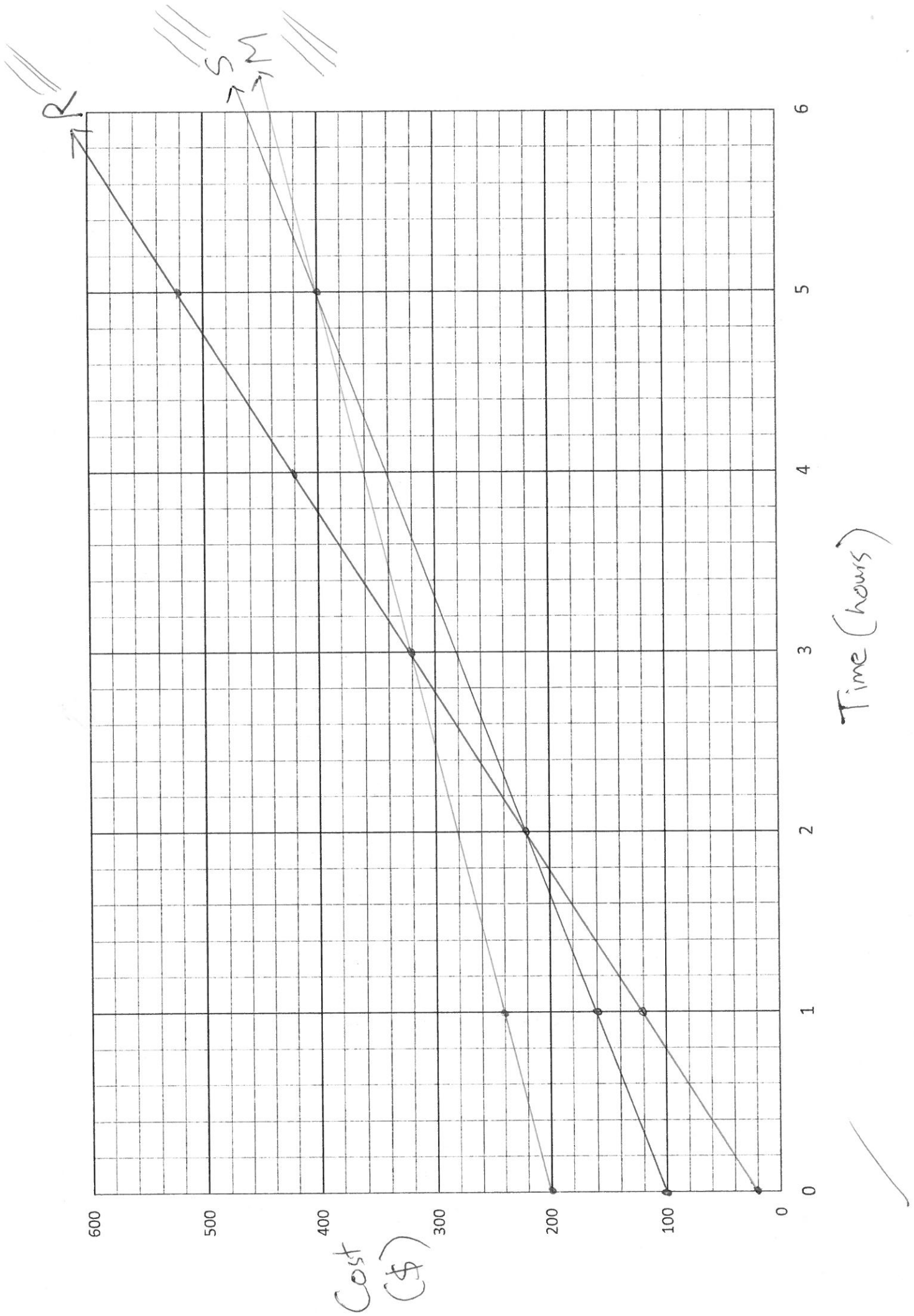
$$C = 40n + 200$$

Steve's Death Metal Band.

N (number of hours)	0	1	2	3	4	5	6
C (cost in \$)	100	160	220	280	340	400	460

Now write a rule you can use to calculate the cost Steve's Death Metal Band.

$$C = 60n + 100$$



2. Plot these three options on the graph paper provided. **Label each option** and the axes. (10 marks)

3. Describe how value for money changes for the three options depending on the length of the disco. (3 marks) Eg Which band is cheapest when?

Rick is the cheapest for the first 2 hrs ✓
Then Steve becomes the cheapest for 3 hrs ✓
After that, Mark will be cheapest ✓

4. Which option is the best value if the disco is to go for 1 hours? (1 mark)

Rick ✓

5. How much will the cheapest music act cost if the disco goes for 4 hours? (1 mark)

\$340 ✓

6. a) The principal has set the times for the disco from 7pm until 10:30pm. Using your graph, find out what the cost would then be for each of the three options. (9 marks)

Rick's Rock Group

\$370 ✓

Magic Mark's DJ

\$340 ✓

Steve's Death Metal

\$310 ✓

- b) Check that these values are correct by substituting N (number of hours) back into your rule to find the cost. Show your working.

Rick's Rock Group

$$C = 100 \times 3.5 + 20$$

$$C = 350 + 20$$

$$C = \$370$$

Magic Mark's DJ

$$C = 40(3.5) + 200$$

$$C = 140 + 200$$

$$C = \$340$$

Steve's Death Metal

$$C = 60(3.5) + 100$$

$$C = 210 + 100$$

$$C = \$310$$

7. If you are given a budget of \$300, how many hours could you hire Magic ^{Mark} Mike's DJ service for? (1 mark)

$$2\frac{1}{2} \text{ hrs} \quad /$$

8. Sales of tickets has resulted in 30 tickets getting bought at \$10 each. Using the total money raised which band would be able to play the longest? Explain your answer. (2 marks)

$$\$10 \times 30 = \$300 \text{ raised} \quad /$$

Steve play the longest /

9. If an additional 15 tickets are sold would this change which group you would choose? Explain your answer. (2 mark)

$$\$10 \times 45 = \$450 \text{ raised}$$

Changes to Mark

10. If I were to pay Steve's Death Metal Band \$1000 to play at my birthday how many hours could I hire them for? Create and solve a linear equation to prove your answer. (3 marks)

$$60n + 100 = 1000$$

$$\begin{array}{r} -100 \end{array} \quad \begin{array}{r} -100 \end{array}$$

$$60n = 900$$

$$\begin{array}{r} \div 60 \end{array} \quad \begin{array}{r} \div 60 \end{array}$$

$$n = 15$$

You could hire them for 15 hrs